Assignment 2

1. **(10 points)** Show that the ridge regression estimates can be obtained by ordinary least squares regression on an augmented data set. We augment the centered matrix $\mathbf{X}$ with p additional rows $\sqrt{\lambda}\ \mathbf{I}$, and augment $\mathbf{y}$ with p zeros. By introducing artificial data having response value zero, the fitting procedure is forced to shrink the coefficients toward zero. This is related to the idea of *hints* due to Abu-Mostafa (1995), where model constraints are implemented by adding artificial data examples that satisfy them.

2. **(30 points)** Recall that a generative classifier estimates $P(\mathbf{X}, Y) = P(Y)P(\mathbf{X}|Y)$, while a discriminative classifier directly estimates $P(Y|\mathbf{X})$. Assume

   (a) Y is a boolean variable following a Bernoulli distribution, with parameter $\pi = P(Y = 1)$ and thus $P(Y = 0) = 1 - \pi$.

   (b) $\mathbf{X} = (X_1, \ldots, X_p)$, where each attribute X_j is a continuous random variable. For each X_j, $P(X_j|Y = k)$ is a Gaussian distribution $N(\mu_{jk}, \sigma_j^2)$, which does **not** depend on k.

   (c) For all $i \neq j$, X_i and X_j are conditionally independent given Y. This us why this type of classifier is called "naive".

   We say this is a specific class of Gaussian naive Bayes classifier because we have made an assumption that the standard derivation σ_j of $P(X_j|Y = k)$ does not depend of k. This is **not** a general assumption for Gaussian naive Bayes classifiers. Show that the form of $P(Y|\mathbf{X})$ derived from the assumptions of a specific class of Gaussian naive Bayes classifiers is precisely the form used by logistic regression. **[10 points]**

   Next, let's make our Gaussian naive Bayes classifiers a little more general by removing the assumption that the standard deviation σ_j of $P(X_j|Y = k)$ does

not depend of k. As a result, for each X_j, $P(X_j|Y = k)$ is a Gaussian distribution $N(\mu_{jk}, \sigma^2_{jk})$, where $j = 1, \ldots, p$, $k = 0, 1$. Is the new form of $P(Y|\mathbf{X})$ implied by this more general Gaussian naive Bayes classifier still the form used by logistic regression? Derive the new form of $P(Y|\mathbf{X})$ to prove your answer. [**10 points**]

Finally, you will implement the Gaussian Naive Bayes classifier and apply it to the spam data (`https://hastie.su.domains/ElemStatLearn/data.html`) in the textbook. [**10 points**]

3. **(30 points)** The MNIST dataset is a famous dataset in computer vision, which is available at `https://www.kaggle.com/datasets/oddrationale/mnist-in-csv`. The data files mnist_train.csv and mnist_test.csv contain gray-scale images of hand-drawn digits, from 0 through 9.

Each image is 28 pixels in height and 28 pixels in width, for a total of 784 pixels in total. Each pixel has a single pixel-value associated with it, indicating the lightness or darkness of that pixel, with higher numbers meaning darker. This pixel-value is an integer between 0 and 255, inclusive.

Let's consider a **binary** classification problem by extracting data only for digits 5 and 8 from both training and test datasets.

Analyze the MNIST data for digits 5 and 8 by using

— a single tree

— bagging

— random forest

— boosting

Compare their performances of these methods based on the misclassification error on the test dataset. You are not allowed to use any information from the

test dataset to produce the classifier. For each method, since we have the same training dataset and the same evaluation criteria, your score for this part will be based on your ranking among all students' submissions.